Fall 2025

ECE 216A Homework 3

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Question 1

a.

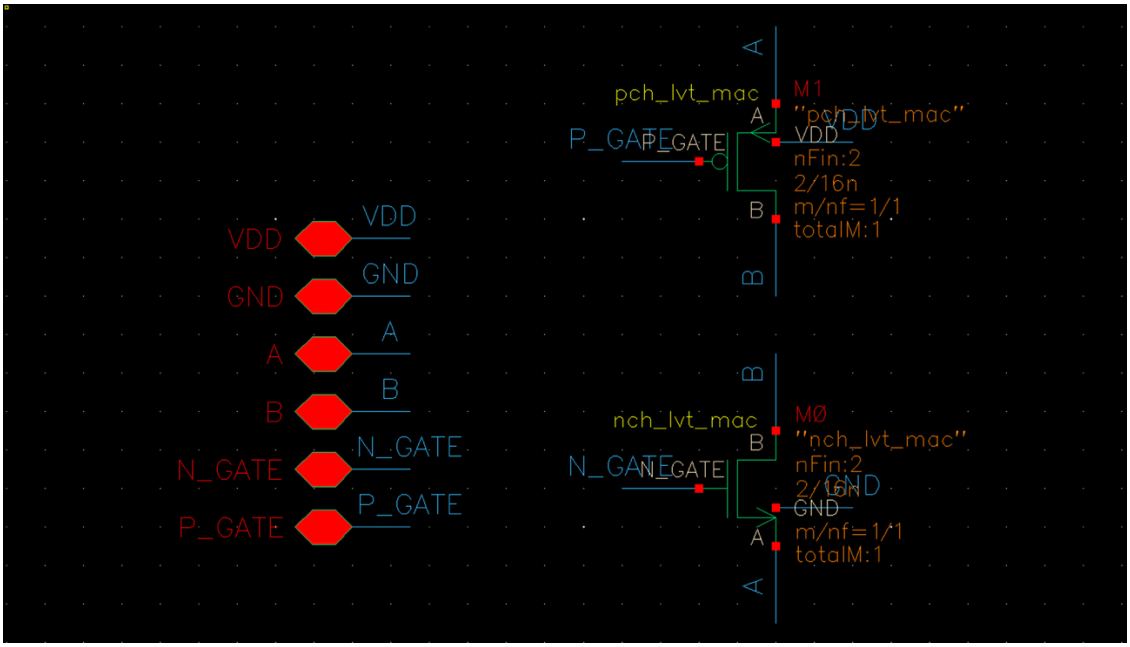



Figure 1. T-Gate Schematic

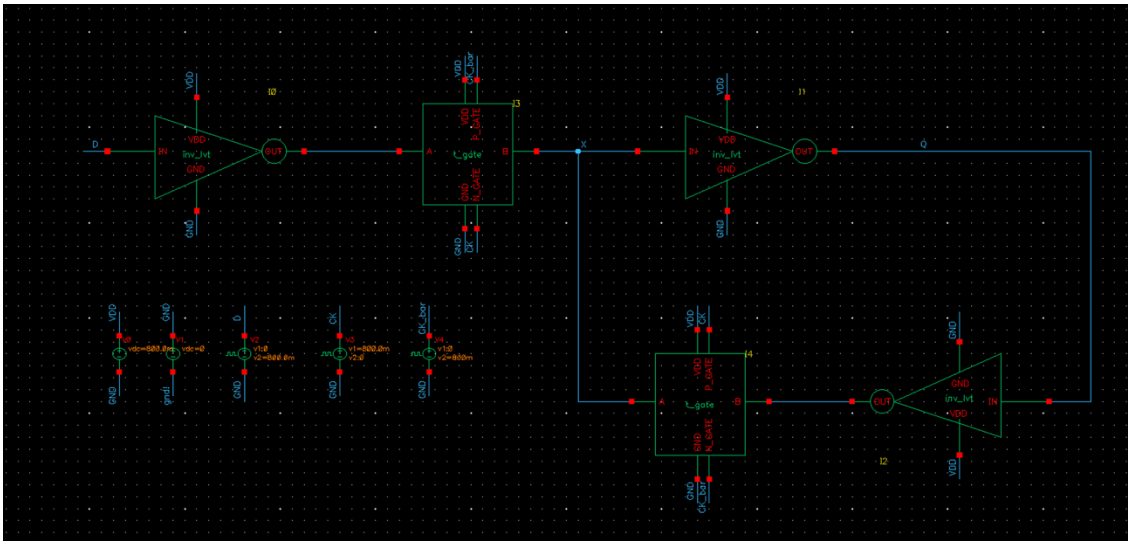


Figure 2. Overall Schematic

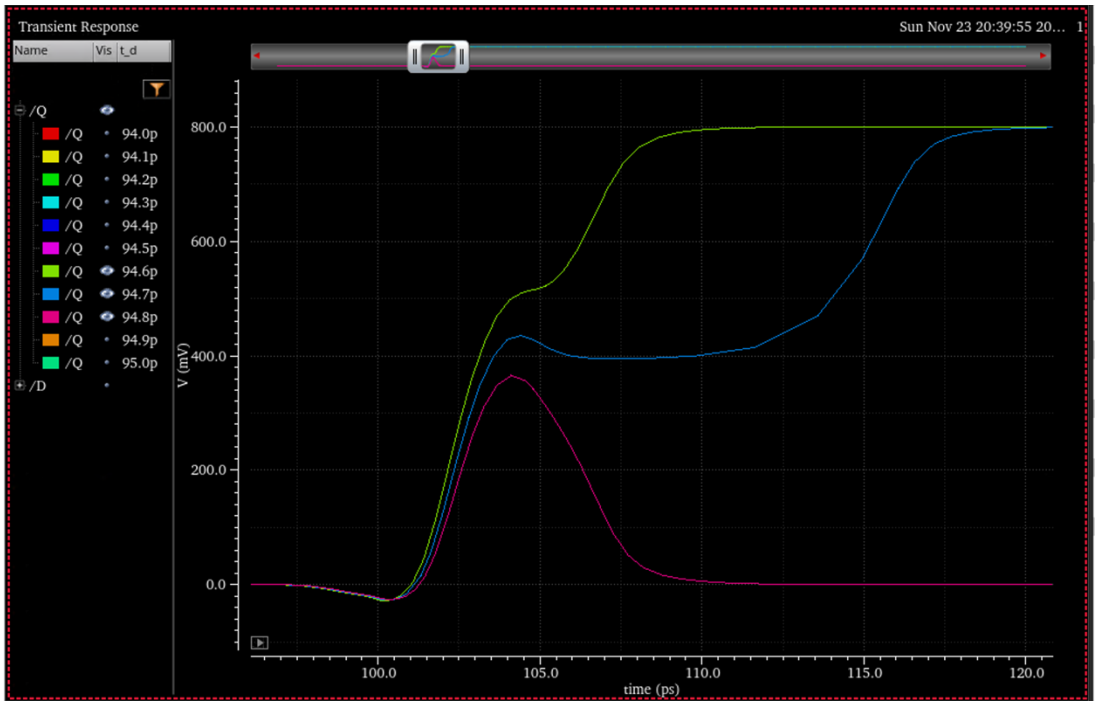



Figure 3. Setup Time Simulation

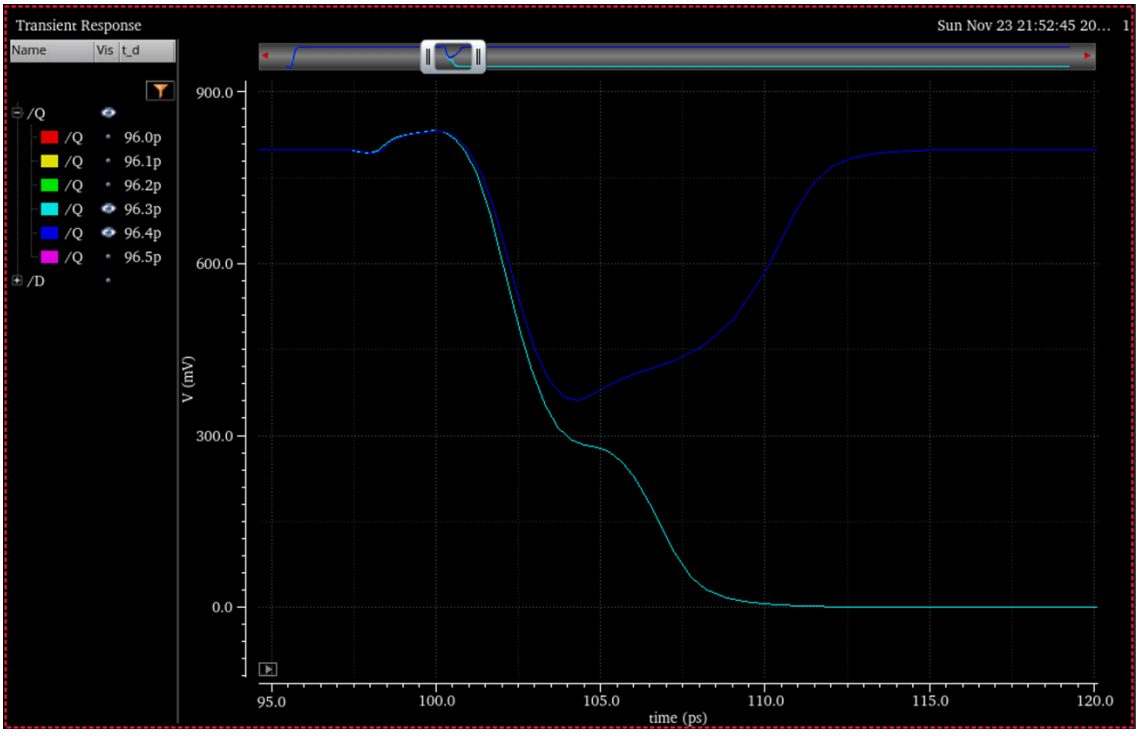



Figure 4. Hold Time Simulation

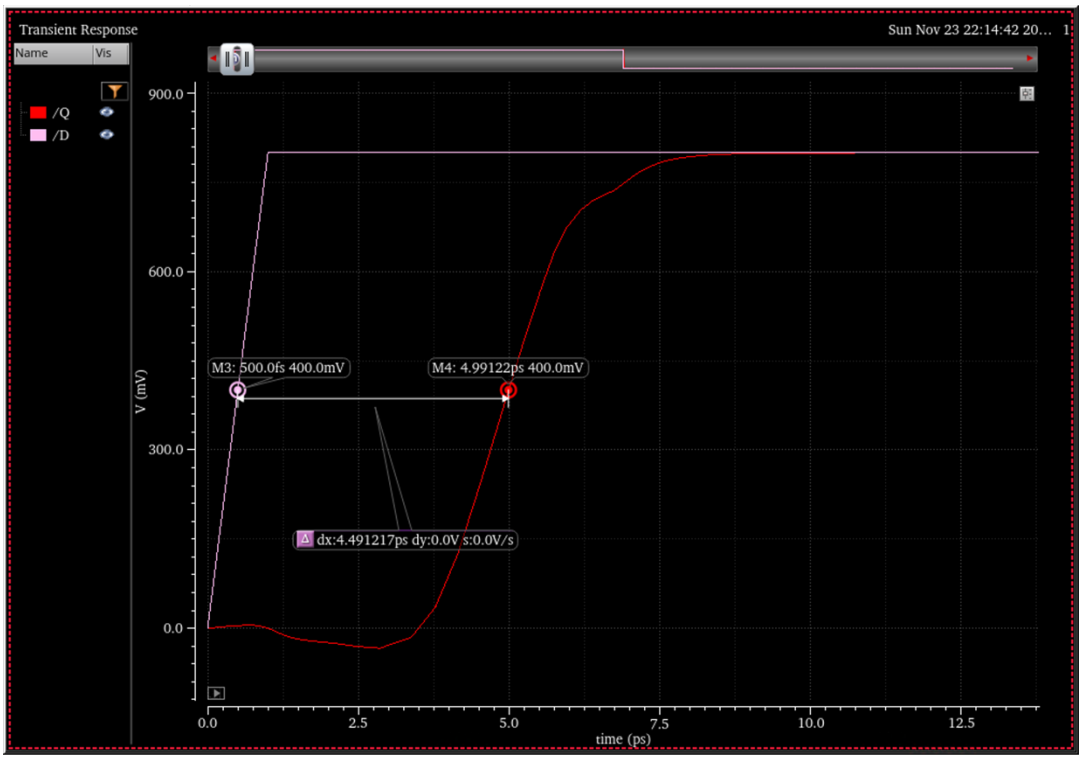



Figure 5. t_dDQ Simulation

t_s = 100 ps – 94.7 ps = 5.3 ps

t_h = 96.4 ps – 100 ps = -3.6 ps

t_dDQ = 4.49 ps

b.

From separate simulations, each INVX0 and passgate has a same delay of. The D to Q path during the transparent phase goes through two inverters and one t-gate, so we expect t_dDQ = 2*t_inv + t_tgate, which matches the simulated value 4.49 ps.

c.

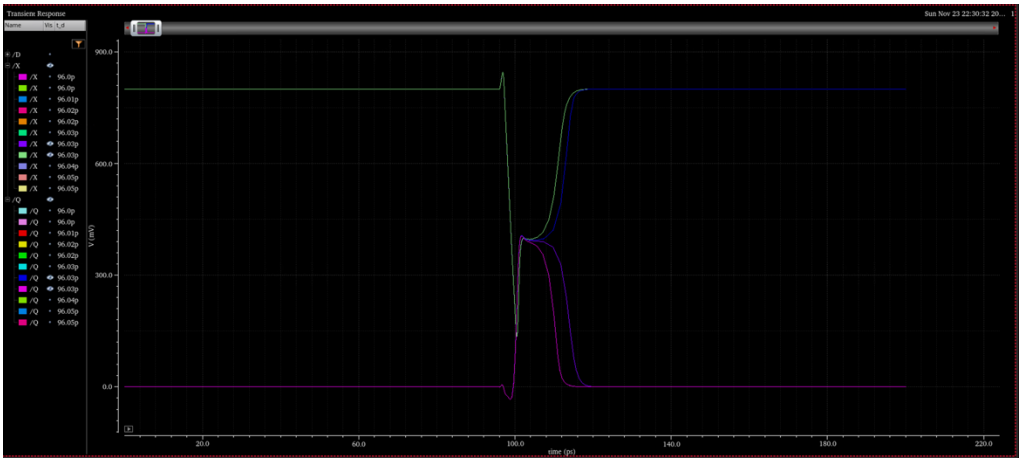



Figure 6. Node X and Q Simulation when Latch in Metastability

2.

$T_{CLK} = 10ns$

$t_{jitter} = 1ns$

$t_{skew} = 1ns$

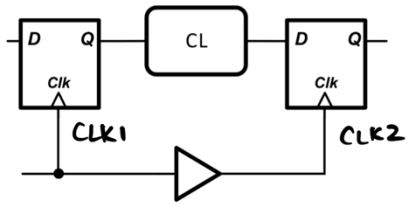
$t_d = 4ns$

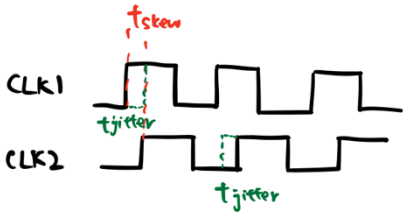
$t_c = 3ns$

$t_{dcq} = t_{ccq} = 2ns$

$t_s = 2ns$

$t_h = 1ns$



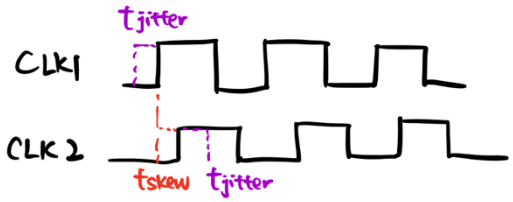

① Check if satisfying cycle time constraint :

$T - 2t_{jitter} + t_{skew} > t_{dcq} + t_d + t_s$

$10ns - 2ns + 1ns > 2ns + 4ns + 2ns$

$9ns > 8ns$

cycle time constraints are satisfied



② check if satisfying hold time constraint:

$$t_{cdq} + t_c - t_{jitter} > t_{skew} + t_{jitter} + t_h$$

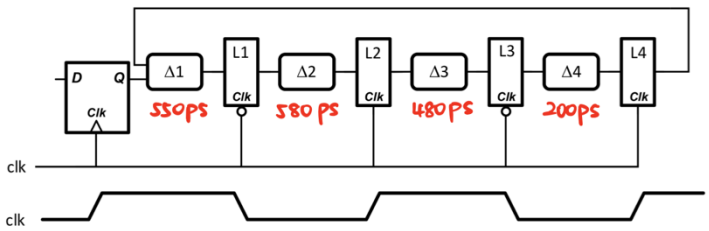
$$2ns + 3ns - 1ns > 1ns + 1ns + 1ns$$

$$4ns > 3ns$$

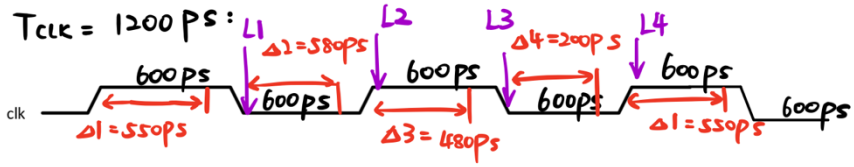
hold time constraints are satisfied.

In summary, there are not time violations.

3. $T_{clk}$ = 1200ps, 1000ps, 800ps

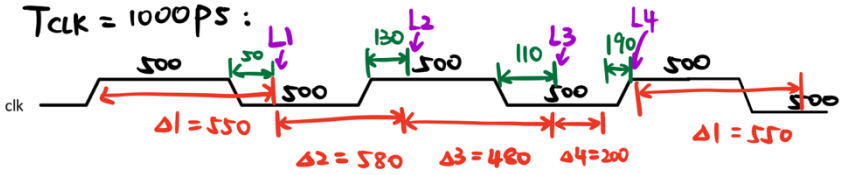



① $T_{CLK}$ = 1200 ps:



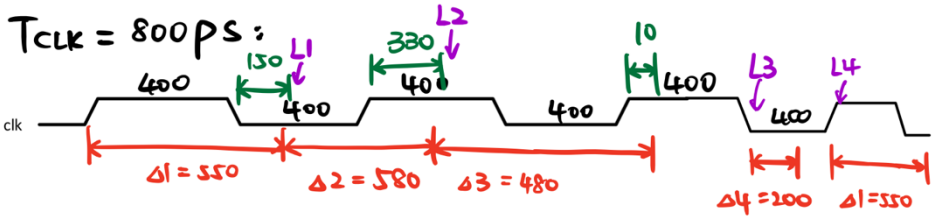

no latches borrow time, no setup violation

② $T_{CLK}$ = 1000ps:




L1, L2, L3 borrow time, no setup violation

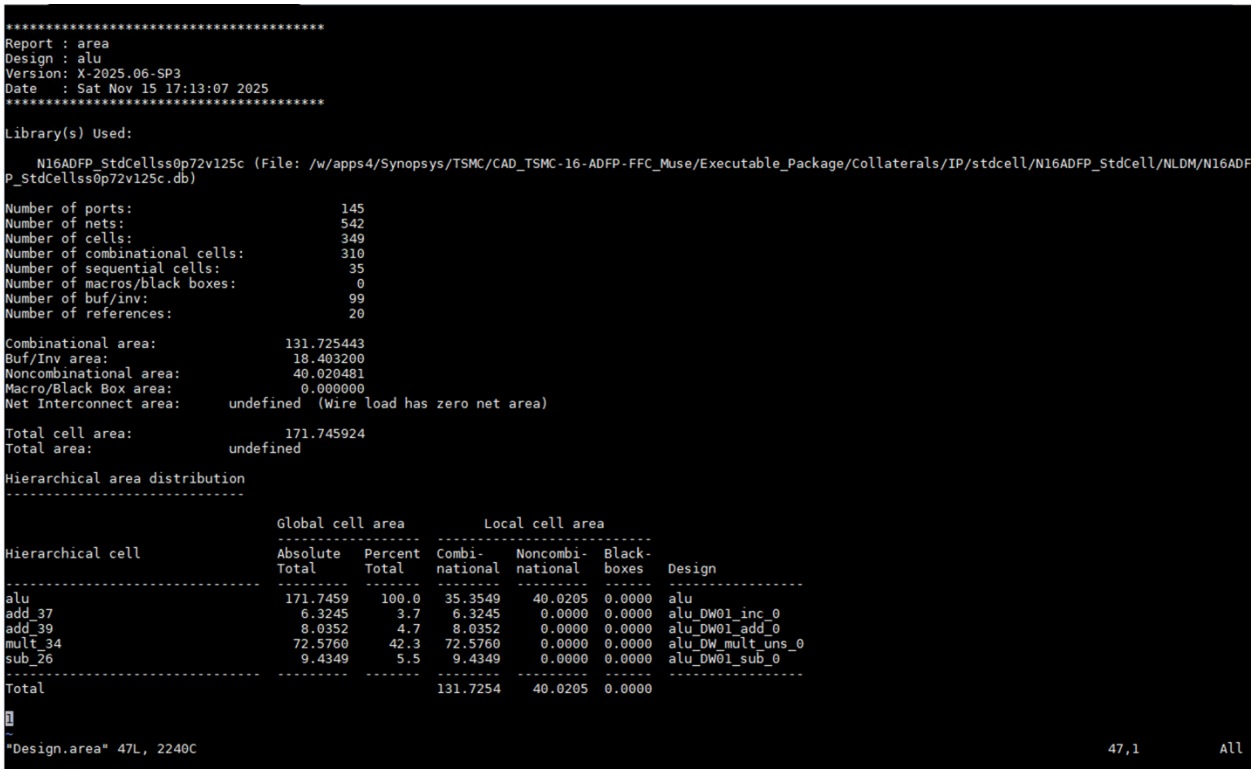
③ $T_{CLK}$ = 800ps:




L1, L2, L3 borrow time, setup violation occurs.

Question 4

a.



```
****************************************
Report : area
Design : alu
Version: X-2025.06-SP3
Date   : Sat Nov 15 17:13:07 2025
****************************************

Library(s) Used:

    N16ADFP_StdCellss0p72v125c (File: /w/apps4/Synopsys/TSMC/CAD_TSMC-16-ADFP-FFC_Muse/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/NLDM/N16ADF
P_StdCellss0p72v125c.db)

Number of ports:                          145
Number of nets:                           542
Number of cells:                          349
Number of combinational cells:            310
Number of sequential cells:                35
Number of macros/black boxes:               0
Number of buf/inv:                         99
Number of references:                      20

Combinational area:               131.725443
Buf/Inv area:                      18.403200
Noncombinational area:             40.020481
Macro/Black Box area:               0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                  171.745924
Total area:                 undefined

Hierarchical area distribution
------------------------------

                                Global cell area          Local cell area
                                ------------------  --------------------------
Hierarchical cell               Absolute   Percent  Combi-     Noncombi-  Black-
                                Total      Total    national   national   boxes   Design
------------------------------  ---------  -------  --------  ---------  ------  -----------------
alu                              171.7459    100.0   35.3549    40.0205  0.0000  alu
add_37                             6.3245      3.7    6.3245     0.0000  0.0000  alu_DW01_inc_0
add_39                             8.0352      4.7    8.0352     0.0000  0.0000  alu_DW01_add_0
mult_34                           72.5760     42.3   72.5760     0.0000  0.0000  alu_DW_mult_uns_0
sub_26                             9.4349      5.5    9.4349     0.0000  0.0000  alu_DW01_sub_0
------------------------------  ---------  -------  --------  ---------  ------  -----------------
Total                                                131.7254    40.0205  0.0000

1
~
"Design.area" 47L, 2240C                                                                47,1          All
```

Figure 7. Area Data for Logic Synthesis

Combinational area is 131.73 $\mu m^2$. Buf/Inv area is 18.4 $\mu m^2$. Non-combinational area is 40.02 $\mu m^2$.

b.

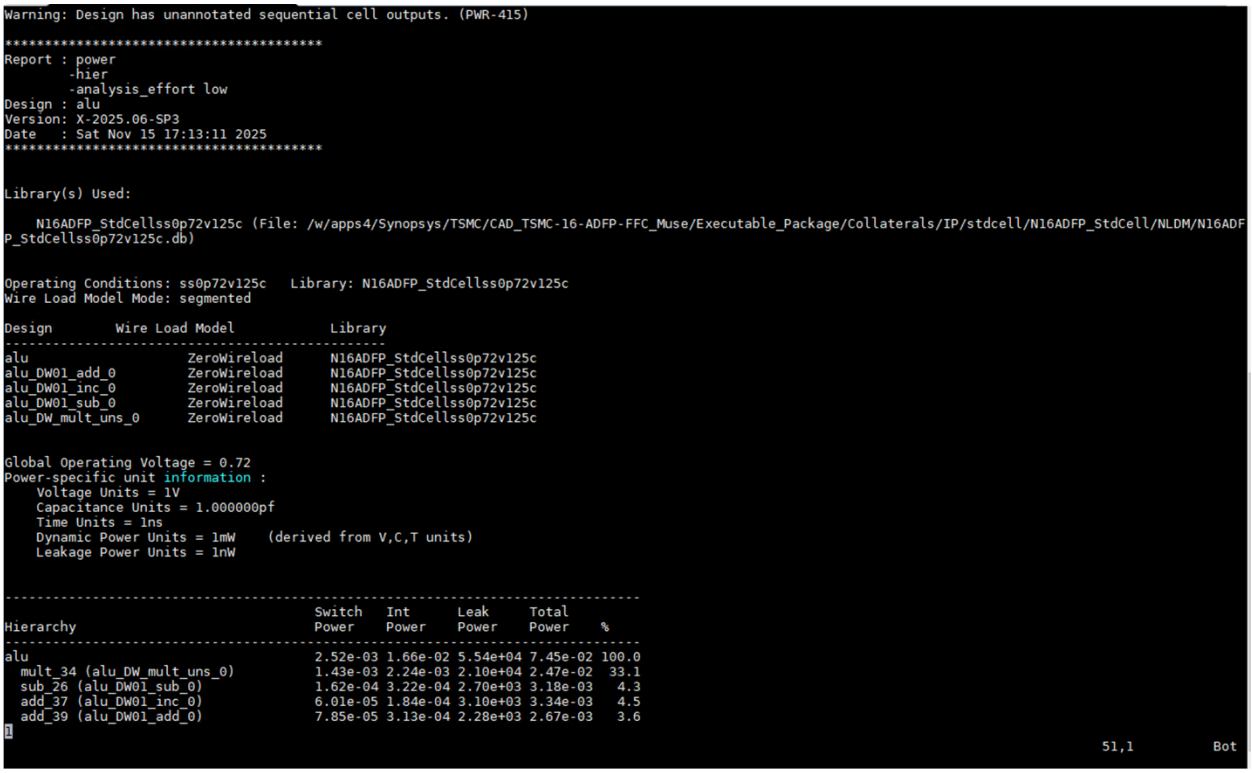



Figure 8. Power Data for Logic Synthesis

Switch power is $2.52\times 10^{-3}\ mW$. Int power is $1.66\times 10^{-2}\ mW$. Leak power is $5.54\times 10^{4}\ nW$. Total power is $7.45\times 10^{-2}\ mW$.

c.

```
Operating Conditions: ff0p88vm40c   Library: N16ADFP_StdCellff0p88vm40c
Wire Load Model Mode: segmented

  Startpoint: Data_A_o_r_reg[0]
              (rising edge-triggered flip-flop clocked by clk_p_i)
  Endpoint: ALU_d2_r_reg[0]
            (rising edge-triggered flip-flop clocked by clk_p_i)
  Path Group: clk_p_i
  Path Type: min

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  alu                ZeroWireload          N16ADFP_StdCellss0p72v125c
  alu_DW_mult_uns_0  ZeroWireload          N16ADFP_StdCellss0p72v125c
  alu_DW01_sub_0     ZeroWireload          N16ADFP_StdCellss0p72v125c
  alu_DW01_add_0     ZeroWireload          N16ADFP_StdCellss0p72v125c

  Point                                                   Incr       Path
  --------------------------------------------------------------------------
  clock clk_p_i (rise edge)                               0.00       0.00
  clock network delay (ideal)                             0.00       0.00
  Data_A_o_r_reg[0]/CP (DFCNQD2BWP16P90LVT)               0.00       0.00 r
  Data_A_o_r_reg[0]/Q (DFCNQD2BWP16P90LVT)                0.04       0.04 f
  U335/ZN (AOI22D1BWP16P90LVT)                            0.01       0.05 r
  U233/Z (CKBD1BWP16P90LVT)                               0.01       0.06 r
  U332/ZN (ND3D1BWP16P90LVT)                              0.01       0.07 f
  U234/Z (CKBD1BWP16P90LVT)                               0.01       0.08 f
  U232/Z (CKBD1BWP16P90LVT)                               0.01       0.09 f
  U231/Z (CKBD1BWP16P90LVT)                               0.01       0.10 f
  U230/Z (CKBD1BWP16P90LVT)                               0.01       0.11 f
  ALU_d2_r_reg[0]/D (DFCNQD2BWP16P90LVT)                  0.00       0.11 f
  data arrival time                                                  0.11

  clock clk_p_i (rise edge)                               0.00       0.00
  clock network delay (ideal)                             0.00       0.00
  clock uncertainty                                       0.10       0.10
  ALU_d2_r_reg[0]/CP (DFCNQD2BWP16P90LVT)                 0.00       0.10 r
  library hold time                                       0.01       0.11
  data required time                                                 0.11
  --------------------------------------------------------------------------
  data required time                                                 0.11
  data arrival time                                                 -0.11
  --------------------------------------------------------------------------
  slack (MET)                                                        0.00
```

Figure 9. Hold Timing Data for Logic Synthesis

The stacks from the 5 worst hold times are 0.

d.

```
****************************************
Report : timing
         -path full
         -delay max
         -nworst 2
         -max_paths 10
Design : alu
Version: X-2025.06-SP3
Date   : Sat Nov 15 17:13:07 2025
****************************************

Operating Conditions: ss0p72v125c   Library: N16ADFP_StdCellss0p72v125c
Wire Load Model Mode: segmented

  Startpoint: ALU_d2_r_reg[9]
              (rising edge-triggered flip-flop clocked by clk_p_i)
  Endpoint: data_o[9] (output port clocked by clk_p_i)
  Path Group: clk_p_i
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  alu                ZeroWireload          N16ADFP_StdCellss0p72v125c

  Point                                                   Incr       Path
  --------------------------------------------------------------------------
  clock clk_p_i (rise edge)                               0.00       0.00
  clock network delay (ideal)                             0.00       0.00
  ALU_d2_r_reg[9]/CP (DFCNQD2BWP16P90LVT)                 0.00       0.00 r
  ALU_d2_r_reg[9]/Q (DFCNQD2BWP16P90LVT)                  0.04       0.04 f
  data_o[9] (out)                                         0.00       0.04 f
  data arrival time                                                  0.04

  clock clk_p_i (rise edge)                               8.00       8.00
  clock network delay (ideal)                             0.00       8.00
  clock uncertainty                                      -0.10       7.90
  output external delay                                  -0.60       7.30
  data required time                                                 7.30
  --------------------------------------------------------------------------
  data required time                                                 7.30
  data arrival time                                                 -0.04
  --------------------------------------------------------------------------
  slack (MET)                                                        7.26
```

Figure 10. Setup Timing Data for Logic Synthesis

The stacks from the 5 worst setup times are 7.26 ns.

5.

a.

```
Point                                                   Incr       Path
-----------------------------------------------------------------------------
clock clk_p_i (rise edge)                               0.00       0.00
clock network delay (propagated)                        0.00       0.00
input external delay                                    0.60       0.60 r
inst_i[0] (in)                                          0.00       0.60 r
Inst_o_r_reg[0]/D (DFCNQND1BWP16P90LVT)                 0.00       0.60 r
data arrival time                                                  0.60

clock clk_p_i (rise edge)                               8.00       8.00
clock network delay (propagated)                        0.00       8.00
Inst_o_r_reg[0]/CP (DFCNQND1BWP16P90LVT)                           8.00 r
clock reconvergence pessimism                           0.00       8.00
clock uncertainty                                      -0.10       7.90
library setup time                                     -0.01       7.89
data required time                                                 7.89
-----------------------------------------------------------------------------
data required time                                                 7.89
data arrival time                                                 -0.60
-----------------------------------------------------------------------------
slack (MET)                                                        7.29


Startpoint: data_b_i[7] (input port clocked by clk_p_i)
Endpoint: Data_B_o_r_reg[7] (rising edge-triggered flip-flop clocked by clk_p_i)
Path Group: clk_p_i
Path Type: max

Point                                                   Incr       Path
-----------------------------------------------------------------------------
clock clk_p_i (rise edge)                               0.00       0.00
clock network delay (propagated)                        0.00       0.00
input external delay                                    0.60       0.60 r
data_b_i[7] (in)                                        0.00       0.60 r
Data_B_o_r_reg[7]/D (DFCNQD2BWP16P90LVT)                0.00       0.60 r
data arrival time                                                  0.60

clock clk_p_i (rise edge)                               8.00       8.00
clock network delay (propagated)                        0.00       8.00
Data_B_o_r_reg[7]/CP (DFCNQD2BWP16P90LVT)                          8.00 r
clock reconvergence pessimism                           0.00       8.00
clock uncertainty                                      -0.10       7.90
library setup time                                     -0.01       7.89
data required time                                                 7.89
-----------------------------------------------------------------------------
data required time                                                 7.89
data arrival time                                                 -0.60
-----------------------------------------------------------------------------
slack (MET)                                                        7.29
```

Figure 11. Setup Timing Data for Logic Synthesis

The stacks for the 2 paths in PrimeTime are 7.29 ns.

b.

```
# UNCOMMENT FOR PART 2
set power_enable_analysis true

###############################################################################
# Of course, ptpx not only gives you accurate static-timing analysis, but also #
# gives you an easy way to model your power. The power estimate can be done in #
# two ways, either you want a averaged power or a data-dependent, cycle-       #
# accurate power analysis based on a particular testvectors. For average       #
# power, you only need to do this:                                             #
#                                                                              #
#    set power_analysis_mode averaged                                          #
#                                                                              #
# For cycle-accurate power (instantaneous peak power for each clock cycle),    #
# you need to do:                                                              #
#                                                                              #
#    set power_analysis_mode time_based                                        #
#    set_power_analysis_options -cycle_accurate_clock clk_p_i                  #
#                                                                              #
###############################################################################

# ADD CODE HERE FOR PART 2
set power_analysis_mode averaged

###############################################################################
# Read the .vcd files. This is a file containing toggling activities of a      #
# particular testvectors (i.e. you send a bunch of test data into your design, #
# and record the switching activities of each of the internal nodes). To get   #
# this file, you need to have (1)Gate-level netlist, including design.vg and   #
# testbench.v; (2)SDF files. To annotate SDF information and also dump the     #
#.vcd file, you just need to create a new initial block in testbench.v and     #
# write the below lines of code. This has ALREADY been done for you in the     #
# provided testbench, but you will have to make these changes yourself in the  #
# project.                                                                     #
#                                                                              #
# initial begin                                                                #
#   $sdf_annotate("Design.sdf", InstanceNameOfYourTestedModule);               #
#   $dumpfile("CTRL_1V.vcd") ;                                                 #
#   $dumpvars ;                                                                #
# end                                                                          #
#                                                                              #
# If everything runs correctly, then your gate-level simulation should have    #
# technology-dependent delay behaviors. After the simulation, you can also get #
# a .vcd file under the same folder you did your gate-level simulation.        #
###############################################################################

# UNCOMMENT FOR PART 2
read_vcd alu.vcd -strip_path alu_tb/alu_0
update_power
report_power -verbos
```

Figure 12. Code for Power Analysis in PrimeTime_PrimePower.tcl

c.

| TB Instruction | Switching Power | Int. Power | Leak Power | Total Power |
|---|---|---|---|---|
| 000 | 4.383µW | 21.26µW | 7.064µW | 32.70µW |
| 001 | 4.306µW | 20.75µW | 7.043µW | 32.10µW |
| 010 | 4.841µW | 23.49µW | 7.094µW | 35.42µW |
| 011 | 2.122µW | 16.05µW | 6.5µW | 24.67µW |
| 100 | 4.313µW | 20.91µW | 7.015µW | 32.24µW |
| 101 | 2.182µW | 16.27µW | 6.531µW | 24.98µW |
| 110 | 4.141µW | 20.12µW | 7.043µW | 31.3µW |
| 111 | 4.295µW | 20.98µW | 7.016µW | 32.29µW |

Table 1. Average Power for Each ALU Operation